

Understanding Service-Oriented Architecture (SOA): A systematic literature review and directions for further investigation

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ABSTRACT

Service-Oriented Architecture (SOA) has emerged as an architectural approach that enhances the service delivery performance of existing traditional systems while still retaining their most important features. This approach, due to its flexibility of adoption, has gained the attention of both academic and business entities, especially in the development of world-leading technologies such as Cloud Computing (CC) and the Internet of Things (IoT). Although many studies have listed the success factors of SOA, a few minor failures have also been reported in the literature. Despite the availability of rich material on SOA, there is a lack of systematic reviews covering the different aspects of the SOA concept in Information Systems (IS) research. Therefore, the central objective of this study is to review existing issues of SOA and share the findings with the academia. Hence, a systematic literature review (SLR) was conducted to analyse existing studies related to SOA and the factors that led to SOA success and failure from 2009 to 2019. To completely cover all SOA-related research in the IS field, a two-stage review protocol that included automatic and manual searching was applied, resulting in 103 primary studies. The articles were categorised into four research themes, namely: SOA Adoption, SOA Concepts, SOA Impact, and SOA Practice. The result shows that the academic research interest on SOA increased recently with most of the articles covering SOA Practice followed by SOA Adoption. Moreover, the findings of this review highlighted SOA Governance, SOA Strategy, Financial Issues and Costs, and Education and Training as the most significant factors of SOA adoption and implementation. Consequently, the outcomes will assist professionals and experts in organisations as well as academic researchers to focus more on these factors for successfully adopting and implementing SOA.

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1. Introduction

In the last few years, Service-Oriented Architecture (SOA) has received increasing attention in line with the move towards tackling challenges associated with the improvement and maintenance of diverse environments [1,2]. To modernise the organisations software system, migrating from a legacy system to an SOA-based system has become a mainstream trend [3]. Recently, many studies have highlighted the benefits of employing SOA in the development of new world-leading technologies such as the Internet of Things (IoT) [4,5] and Cloud Computing (CC) [6,7] and microservices [8]. This is because SOA offers flexible integration and service reusability due to its service-based modular architecture [9]. SOA also offers transparency because it encapsulates several applications and data sources in the form of a black box. In this way, an integrated pool of Information Technology (IT) resources could still be accessible despite the existence of diverse technologies, language codes, functionalities, and platforms.

In regard to industrial sectors, so far, SOA has proven to be a key paradigm in numerous industries such as banking, healthcare, transport, etc. Along with its many benefits, some of the identified primary studies have revealed that organisations are unable to realise the full benefits of SOA adoption because of several reasons. Among the many obstacles, a lack of knowledge and information about the critical success factors (CSFs) for SOA adoption and implementation were the key causes of failure [10,11]. However, there is no detailed systematic study that has shared the critical influential criteria for successful SOA adoption and implementation. To fill this gap, we believe that it is vital to investigate the significant factors of SOA adoption in organisations because an understanding of these factors would help these organisations maximise the benefits of SOA implementation. To investigate and analyse this topic in more detail, this study focuses on answering the following research questions:

RQ1. What are the primary studies on SOA in the Information Systems (IS) domain and which research themes have been addressed in the literature?

RQ2. What theoretical frameworks and models are emphasised in current SOA research?

RQ3. What are the most influential factors affecting SOA adoption/implementation in organisations?

RQ4. What are the gaps, limitations, and future work recommendations in current SOA research?

To answer these questions, the researchers conducted an in-depth investigation of 103 studies. This study is aimed towards practitioners that are in the process of implementing SOA in their organisations. Moreover, the findings of this study contribute to the existing body of work by providing a list of the most significant factors affecting SOA implementation extracted from case studies and surveys. The rest of this review article is laid out as follows: Section 2 introduces SOA based on the existing literature; Section 3 explains the review method used and how this research was carried out; Section 4 provides the SLR results; Section 5 outlines the result of the research questions; and Section 6 presents the conclusion to this systematic review.

2. Service-oriented architecture

Many researchers have defined SOA based on different perspectives (such as technology, business, and architecture), so there is no exact definition for this term. SOA is not a technology, a product, or a swift solution that could tackle the complexity of IT. Moreover, SOA does not guarantee that it can address all existing IT/IS challenges. Nevertheless, it could be said that SOA is a concept that could be advantageously used in business, IT, IS, and enterprises as a whole [12]. Table 1 presents the different descriptions of SOA according to previous studies.

All the definitions of SOA cited in Table 1 is based on different perspectives. Although none are incorrect, most scholars agree that SOA can be defined as an architectural concept that promotes loose coupling, reusability, interoperability, agility, efficiency, with a focus on breaking each business process into smaller blocks of tasks and functions such as services. These services are well defined and organised and serve as the independent units of standard business functionality that are connected to each other to make a unified business process [13–15,17,20,23,24,26–28]. The OASIS Group [22] and the SOA Manifesto [23] defined SOA as a paradigm while Bieberstein et al. [16] and Marks and Bell [12] defined SOA based on both business and technology perspectives. Therefore, researchers and practitioners have adopted the SOA definition based on both perspectives, as both fit the context of this study.

Many scholars reported that the combination of SOA with other technologies could provide more benefits for organisations. For example, Katsikogiannis et al. [29] emphasised that the integration of SOA and policy-based management could result in the rapid development of new functions, as well as the ability to scale operations and meet capacity levels and various demands even with a large community of distributed objects. In a study, Tang et al. [30] indicated that SOA could be integrated with new medical technologies, intelligent agents, wireless technologies, radio-frequency identification (RFID), and operational Internet procedures to establish new automations for the business process. In another study, Gupta et al. [5] stated that IoT requirements could fulfil by the SOA approach since SOA technology can provide performance measurement, security attack detection, and business intelligence. Moreover, Alsaryrah et al. [4] stressed that SOA has recognised as the chief enabler of IoT. They explained that the combination of IoT services applying SOA fosters the production of value-added and complex IoT applications by integrating atomic services to deliver novel capabilities and utilities.

Furthermore, the benefits of adopting SOA in organisations include reduced costs, a greater return on investment (ROI), the reuse and integration of services and traditional systems, reduced time to market, and improved alignment between business and IT [11,31,32].

Moreover, Yoon and Carter [33] highlighted the importance of business and IT alignment along with other critical success in their analysis of multiple case studies. The findings of their empirical study showed two main benefits of SOA implementation: (i) improved business agility and (ii) decreased cost. The first could be achieved through the quick response of IT to market change and customer demand, the ease of integrating systems, the increased alignment between IT and business, besides better customer service and better data flow. Meanwhile, lower costs could be achieved by decreasing the cost/time of application development, decreasing operational costs, and reusing existing applications and functions [33].

Table 1
SOA definitions.

No	Reference	Definition
1	Booth et al. [13]	"SOA is a form of distributed systems architecture that is typically characterised by the following properties: Logic view, Message orientation, Description orientation, Granularity, Network orientation Platform neutral."
2	Natis [14]	"A software architecture that starts with an interface definition and builds the entire application topology as a topology of interfaces, interface implementations and interface calls. SOA is a relation of services and service consumers, both software modules are large enough to represent a complete business function."
3	Lublinsky [15]	"SOA can be defined as an architectural style promoting the concept of business-aligned enterprise service as the fundamental unit of designing, building, and composing enterprise business solutions."
4	Bieberstein et al. [16]	"SOA is a framework for integrating business processes and supporting IT infrastructure as secure, standardised components – services – that can be reused and combined to address changing business priorities. SOA is an enterprise-wide IT architecture that promotes loose coupling, reuse, and interoperability between systems."
5	Krafzig et al. [17]	"SOA is a software architecture that is based on the key concepts of an application front-end, service, service repository, and service bus. A service consists of a contract, one or more interfaces, and an implementation."
6	Dehne and DiMare [18]	"Service-Oriented Architecture (SOA) is a style of developing and integrating software. It involves breaking an application down into common, repeatable services that can be used by other applications, both internal and external, in an organisation-independent of the applications and computing platforms on which the business and its partners rely."
7	Marks and Bell [12]	"SOA is a conceptual business architecture where business functionality, or application logic, is made available to SOA users, or consumers, as shared, reusable services on an IT network. Services in an SOA are modules of business or application functionality with exposed interfaces, and are invoked by messages."
8	Erl [19,20]	"SOA is a form of technology architecture that adheres to the principles of service-orientation. When realised through the Web services technology platform, SOA establishes the potential to support and promote these principles throughout the business process and automation domains of an enterprise." "SOA establishes an architectural model that aims to enhance the efficiency, agility, and productivity of an enterprise by positioning services as the primary means through which solution logic is represented in support of the realisation of strategic goals associated with service-oriented computing."
9	James McGovern et al. [21]	"SOA is a component-based software module that provides services to other modules."
10	MacKenzie et al. [22]	"Service-Oriented Architecture (SOA) is a paradigm for organising and utilising distributed capabilities that may be under the control of different ownership domains."
11	Arsanjani et al. [23]	"Service orientation is a paradigm that frames what you do. Service-Oriented architecture (SOA) is a type of architecture that results from applying service orientation. We have been applying service orientation to help organisations consistently deliver sustainable business value, with increased agility and cost effectiveness, in line with changing business needs."
12	Haren [24]	"Service-Oriented Architecture (SOA) is an architectural style that supports service-orientation. Service-orientation is a way of thinking in terms of services and service-based development and the outcomes of services."
13	Wheller [25]	"Service-Oriented Architecture is a business environment style of design that guides and encourages the creation of loosely coupled business services that are inter-operable and technology-independent, enabling increased business function and flexibility."

3. Review method

The systematic approach outlined in [34] was employed to answer the research questions posited in this study. As stated by Kitchenham and Charters [35], a systematic review is a procedure of identifying, evaluating, and interpreting existing studies associated with a precise question, a particular topic, or an interesting phenomenon. Many researchers have emphasised on the importance of conducting SLR in various IS contexts [36–39]. The major reasons for conducting a systematic review are: (a) to summarise the available evidence of advantages and inhibitions of a technology, (b) to identify the existing gaps and problems in current studies to recommend further examination for future works, and (c) to provide a framework based on an in-depth understanding of the subject to address new research activities properly [35].

These reasons are the main motivations for conducting the current systematic review. The researcher applied the guidelines of Kitchenham and Charters [35] for this review. An SLR consists of three main steps i.e. planning, conducting, and reporting. Each

step involves some other specific tasks and activities such as the identification of the research questions, the development of a review protocol, the identification of the inclusion and exclusion criteria, the selection of the main studies based on a search strategy, data extraction, data synthesis, and finally, the writing of the main report [35]. These steps are explained in more detail in the following sections.

3.1. Review protocol

After defining the research questions, the second key step in a systematic review is to define the review protocol and the methods for conducting the systematic review with the goal of decreasing researcher bias [40]. The review protocol of this study included defining the research questions, developing the search strategy, applying a stepwise selection of the main studies, defining the inclusion and exclusion criteria, data extraction, synthesis and results, and reporting. Fig. 1 presents the review protocol applied for this systematic study.

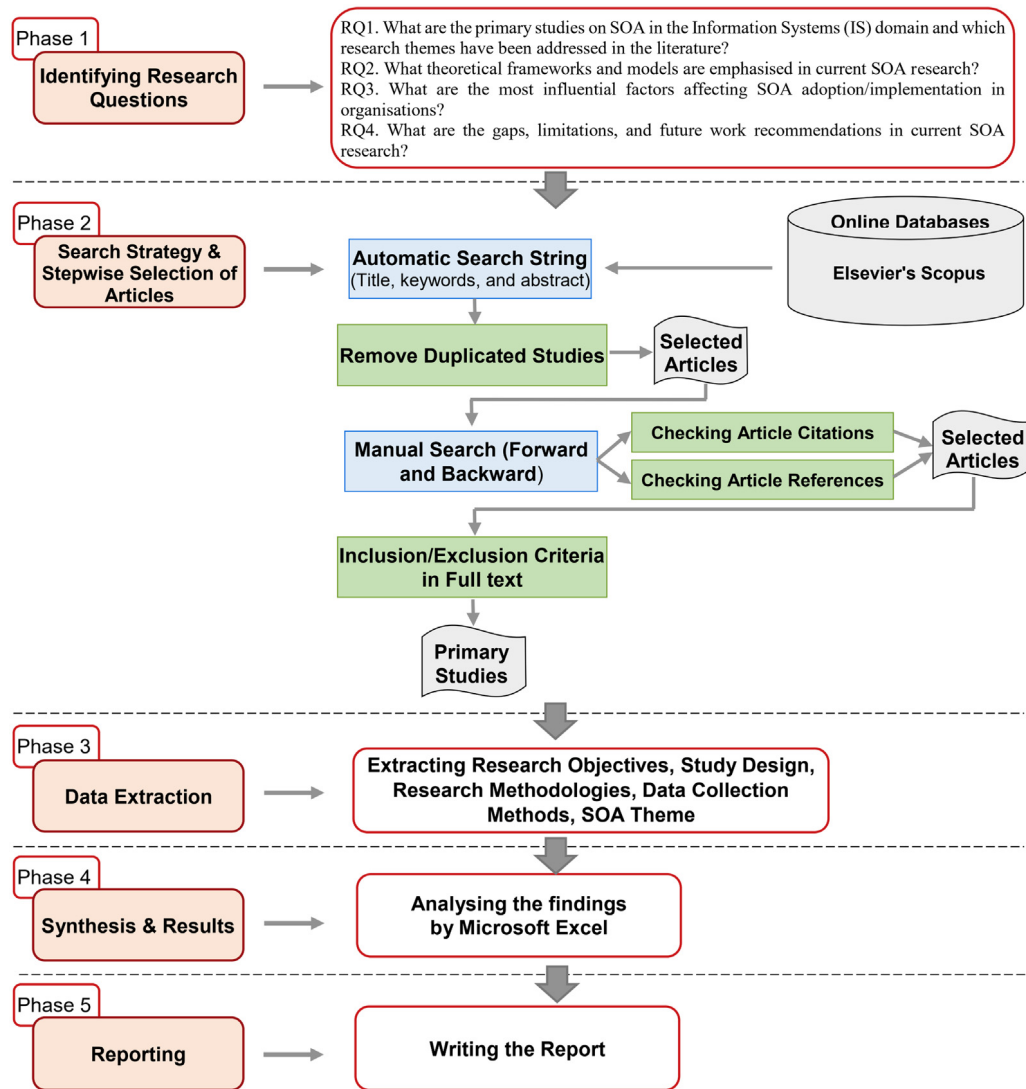


Fig. 1. Review protocol.

3.2. Inclusion and exclusion criteria

Academics have defined some inclusion and exclusion criteria to assist researchers to determine the relevance of selected articles against the objectives of the study. Since the main goal of this study is to review the significant effects of implementing SOA in organisations, this research focused on articles published in the English language from 2009 to the end of November 2019. The year 2009 was selected as the initial year for the search protocol, as the subject of this paper is similar to that of a previous work conducted in 2009 [41]. The past work also aimed to achieve an in-depth understanding of the SOA concept in IS research. The inclusion and exclusion criteria for this review are presented in Table 2.

3.3. Search strategy

This study followed the two-step (automatic and manual) search strategy used in [42]. To accomplish the automatic search strategy, a search string was defined based on the keywords used in SOA research. According to the research questions, the main keywords used are “SOA”, “Information Systems”, and “influential factors”. The operators “OR” and “AND” were used to connect the primary keywords, the synonyms, and some related key terms.

Table 2
Research inclusion and exclusion criteria.

Criteria	Principle
Inclusion	Papers published between 2009 and 2019
	Full text
	Peer-reviewed studies
	Papers focusing on SOA issues
Exclusion	Papers that answer the defined research questions
	Papers not in the English language
	Papers less than 3 pages
	Papers out of the IS scope
	Non-peer reviewed studies

Then, after numerous tests, the search string below was chosen. It should be noted that the researchers used the keywords by shifting the word positions or deleting some of the words during each search process to obtain the most relevant papers. The subsequent search string used were: (SOA or “Service-Oriented architecture”, or “Web Services”) AND (“Information Systems” or “IS”) AND (“adoption” or “migration”, or “legacy systems”) AND (“critical success factors” or CSF or “success factors” or “significant factors” or “influential factors”). To execute these keywords, this study used Elsevier’s Scopus database, which is the

Table 3
The extracted data from primary studies.

Extracted data	Description
Study ID	A unique identity for each primary study.
Author(s)	The name of the authors.
Publication year	The year in which the paper was published.
Title of study	The title of the paper that is visible in the searching step.
Source type	Journal, conference proceeding, workshop, and book chapter.
Country	The place where the empirical studies were conducted.
Subject	The representative group (subject) of data collection in empirical studies.
Methodology	Design science, quantitative, qualitative, mixed method, etc.
Research strategy	Case study, survey, experiment, action research, archival research, etc.
Data collection method	Interview, questionnaire, observation, secondary data, etc.
Theory/Framework	The theory or framework the paper adopted such as Diffusion of Innovation (DOI) theory.
SOA theme	The category of the primary study such as: impact, adoption, concept, or practice.
Objective	The main objective of the papers.
Findings/Factors	The factors or findings of the empirical studies.
Future work	The future works suggested in the primary articles.



Fig. 2. Weighted focus cloud generated by NVivo 10 software.

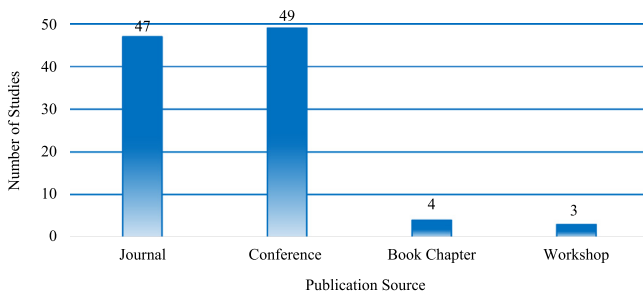


Fig. 3. Distribution of primary studies based on publication source.

largest database for peer-reviewed international conferences and journals in the field of IT.

3.4. Study selection process

In this section, the most relevant studies were identified according to the objectives of the current systematic review. In the first step of the search process, the search strategy defined in Section 3.2 was applied, i.e. automatic search step was used, resulting in 670 articles. In the next stage, the titles, the abstracts, and the conclusion sections of all the collected papers were reviewed and related papers were chosen as the first of the selected studies (143 papers). After removing duplicate papers using Mendeley Software, 131 papers were selected as the secondary selection. According to Levy and Ellis [44], many researchers fall into the

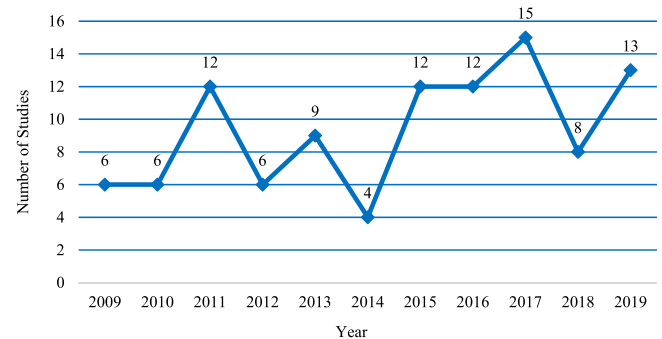


Fig. 4. A temporal view of the primary studies.

trap of a limited literature range, as they only rely on the keywords and buzzwords in the process of searching. To avoid these issues, this study applied the forward and backward technique in the manual searching step. The forward and backward searching method was put forward by Webster and Watson [45] for the first time in their review of the IS literature. The forward search technique involves tracing the papers cited in the selected studies while the backward search technique entails reviewing all the references of each selected study [44,45]. To conduct the forward search method, the Google Scholar search engine was used. After applying the forward and backward methods, a total of 148 papers were selected. Thereafter, full-text scanning was done considering the inclusion and exclusion criteria, resulting in the elimination of 45 papers. Finally, 103 papers were left. These were the primary studies used for this systematic review. These primary studies are presented in [Appendix A](#).

4. Data extraction and synthesis

This section explains the creation of a data extraction form to derive accurate information from the selected articles. In this step, each primary study was read carefully and the relevant data extracted using Mendeley Software and Microsoft Excel spread sheets. The extracted data for this systematic review were input into the Microsoft Excel spread sheet based on the following columns: Study ID, Author(s), and Publication Year, Title of Study, Citation, Online Database, Source Type, Country, Industry, Source Name, Subject, Methodology, Research Strategy, Data Collection Method, Theory/Framework, SOA Theme, Objective, Findings/Factors, and Future Work. These items were included based on the defined research questions and the objectives of the study. The description of these items is presented in [Table 3](#).

As a quick overview, [Fig. 2](#) presents a weighted topic cloud, which was generated by NVivo 10 software. This figure shows the

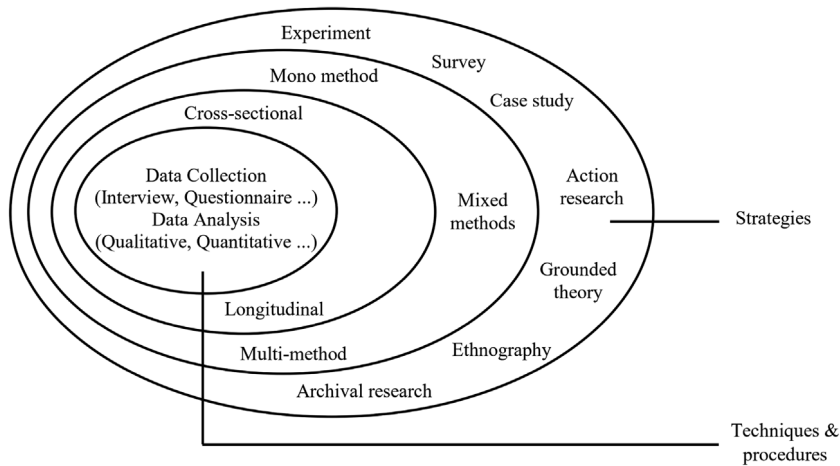


Fig. 5. The research 'onion' [43].

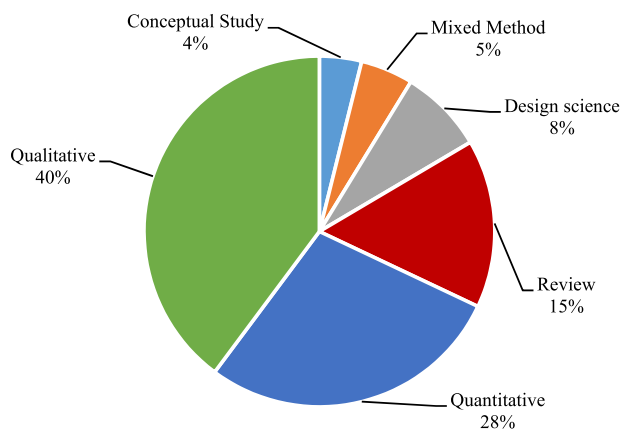


Fig. 6. Distribution of research methodologies.

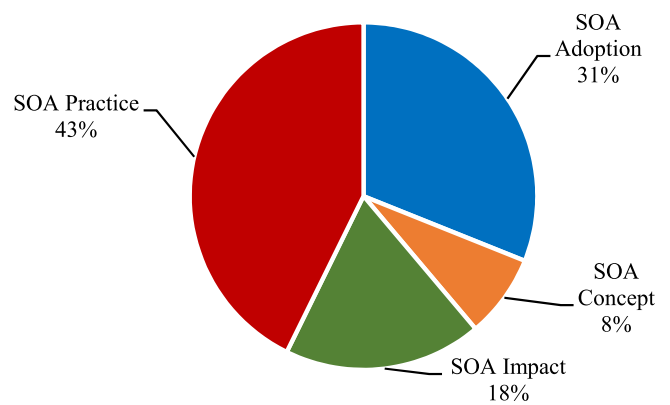


Fig. 8. The distribution of primary studies based on SOA themes.

most frequently used terms in the titles and the abstracts of the primary studies in this review. From Fig. 2, it can be concluded that most of the primary studies discussed service, business, adoption, implementation, organisation, and governance issues.

4.1. An overview of the publication sources

Fig. 3 presents the publication sources of the primary studies. As can be observed, from the total of 103 primary studies, 49 papers were published in conferences followed by 47 papers in journals, 3 papers in workshops, and 4 papers in book chapters.

The full name of the Journals, the Conferences, and the Books is presented in Appendix B.

The majority of the primary studies had selected from the eminent conferences and reliable and high impact factor journals. Most of the reviewed papers were published in Springer (24 Papers) followed by IEEE (22 papers), AISeL (12 papers), and Emerald Insight (8 papers). The remaining studies were published in Science Direct, Taylor & Francis, IGI Global, Inderscience, Wiley OL, SAGE, and ACM DL. Moreover, 15 papers had published in the less renowned publishers. However, their qualifications guaranteed since they have indexed in Elsevier's Scopus.

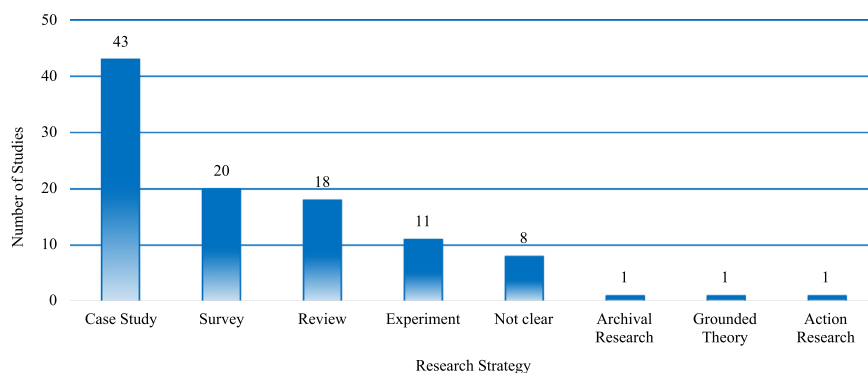


Fig. 7. Distribution of research strategies.

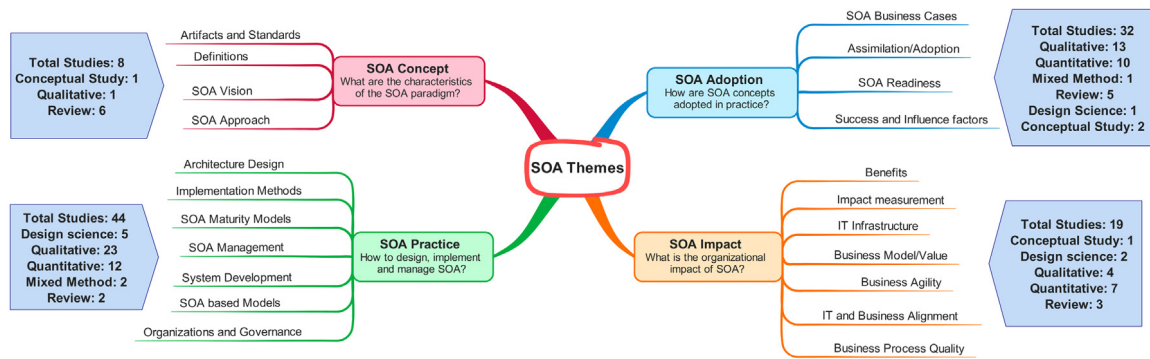


Fig. 9. A mind map of the SOA research themes and research methodologies.

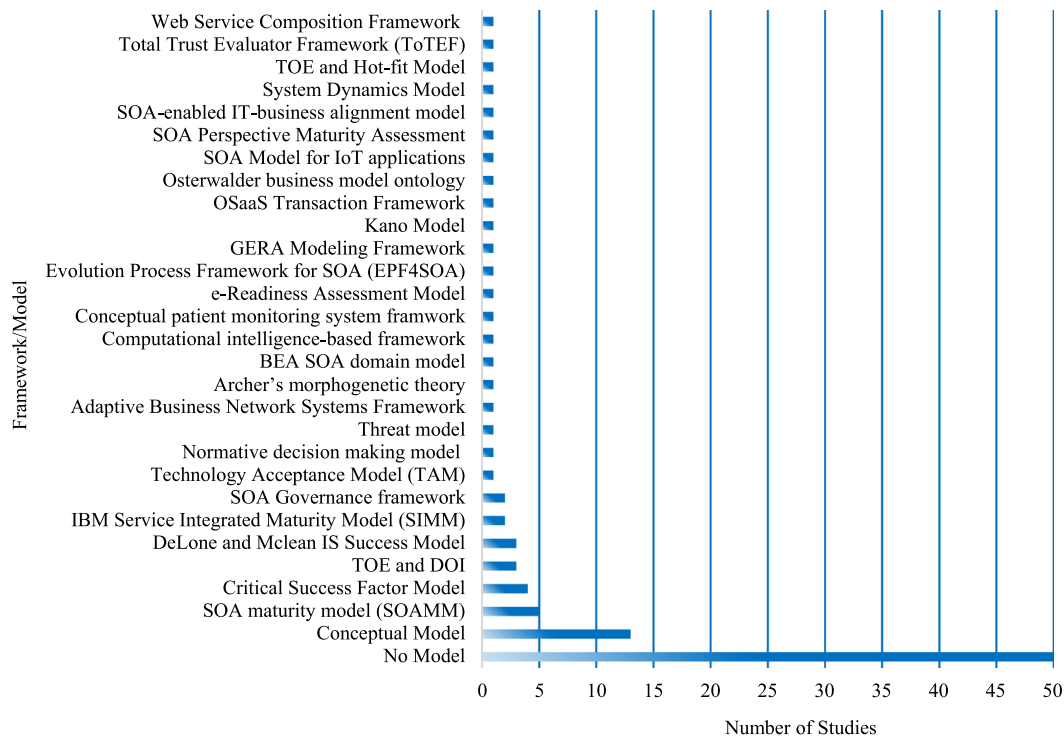


Fig. 10. Theoretical frameworks and models used in the primary studies.

4.2. Temporal view of the publication

As mentioned before, in this systematic review, research papers published between 2009 and 2019 were selected. Fig. 4 presents the distribution of the primary studies during this period. Most articles were published in 2017 (15 papers) followed by 2019 with 13 papers. Moreover, 12 papers were published in each of the years 2016, 2015, and 2011. Out of the 103 published papers, 4 studies were released in 2014, 9 studies in 2013, 8 studies in 2018, and 6 studies in 2009, 2010, and 2012 each.

4.3. Research methods

In many studies, the terms 'research methodology' and 'research method' have been used interchangeably. Saunders et al. [43] defined 'research method' as the techniques and processes employed to obtain and analyse data. The authors stated that this definition includes both the data collection methods (interview, questionnaire, etc.) and the data analysis techniques and procedures (Qualitative, Quantitative, etc.). However, this study used the term 'research methodology' to refer to the techniques

and procedures for analysing data and the term 'data collection method' to refer to the techniques for collecting data. Fig. 5 presents the research 'onion' proposed by Saunders et al. [43].

Fig. 6 presents the methodologies that were adopted by the selected studies. The majority of the primary studies applied a qualitative methodology while the rest adopted the quantitative, design science, or mixed method. This figure illustrates that 40% (41 papers) of the studies were qualitative while 28% (29 papers) were quantitative and sixteen studies were review papers. This distribution showed that eight studies used the design science methodology, four were conceptual studies, and five applied the mixed method (qualitative and quantitative).

As depicted in Fig. 7, the majority of the primary studies employed a case study strategy (43 papers) while 20 articles used a survey strategy, and 18 articles were review study. Out of the 43 case study papers, 18 articles applied multiple case studies while 25 articles used a single case study. Besides, 11 articles employed an experimental research strategy, and one study in each action research, archival research, and grounded theory research strategies. The research strategy of the remaining papers was unclear.

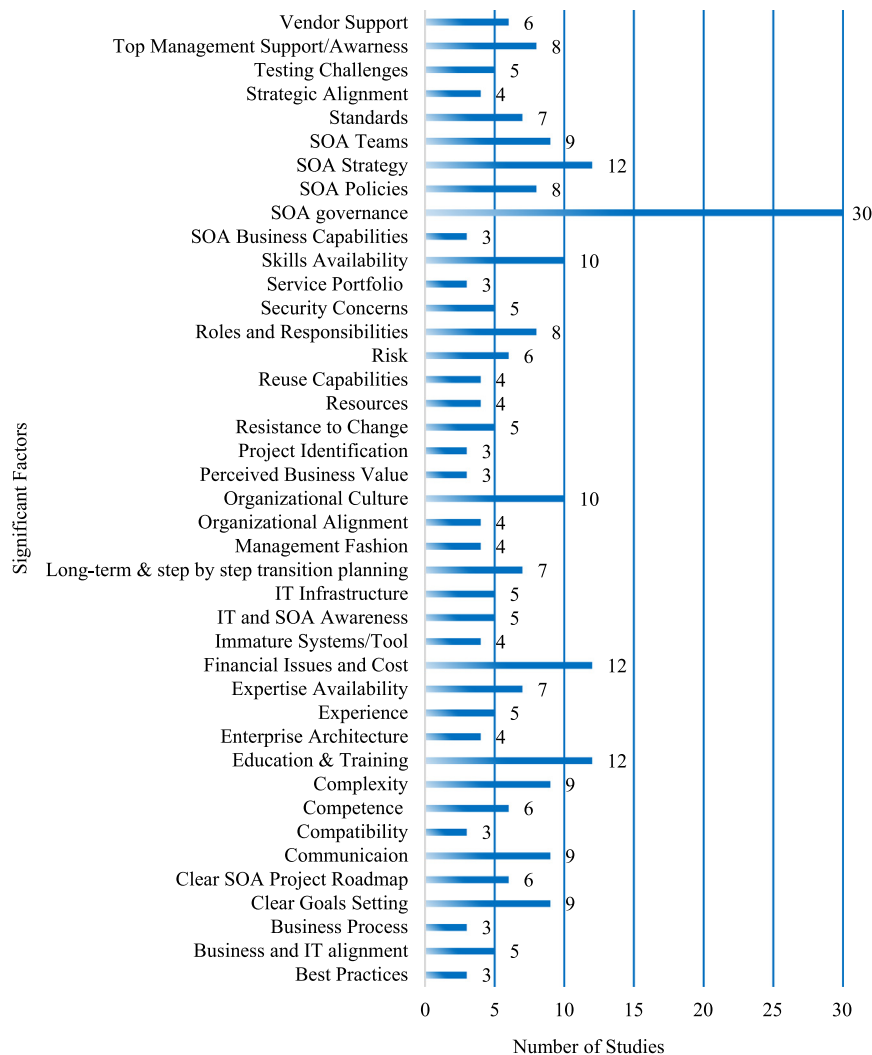


Fig. 11. The most influential factors affecting SOA implementation as extracted from the primary studies.

Table 4

SOA research themes and descriptions.

SOA theme	Description
Concept	Articles that focus on describing and analysing the phenomena of SOA and answer this question: What are the features of the SOA paradigm?
Adoption	Studies in this group focus on the patterns and factors that influence on SOA readiness, adoption, and implementation and answer the following question: How are SOA concepts adopted in practice?
Impact	Include research articles that concentrate on describing and measuring the impact of SOA on organisations and answer this question: What is the organisational impact of SOA?
Practice	Studies that specify how enterprises apply SOA concept from the practitioner perspective and answer this question: How to design, manage, and implement SOA in organisations?

5. Research Question (RQ) results

As mentioned in Section 1, this systematic review aimed to answer some research questions on SOA from the IS perspective. This section discusses the answers to each research question.

RQ (1): What are the primary studies on SOA in the IS domain and which research themes have been addressed in the literature?

Liang and Turban [46] defined the term 'research theme' as the main and central topic that a research study intends to explore. Following a literature review carried out by Viering et al. [41], this study categorised the primary studies into four groups, namely,

SOA Adoption, SOA Impact, SOA Concept, and SOA Practice. Table 4 lists the descriptions of these themes while Appendix A presents the SOA academic studies that have been classified into these themes.

In conclusion, of the 103 studies on SOA, 44 (43%) reported on SOA Practice while 32 articles (31%) focused on SOA Adoption. Moreover, 19 articles (18%) concentrated on the SOA Impact in the organisations, and only 8 studies (8%) reported on the SOA Concept. The overall results are shown in Fig. 8.

Fig. 9 presents a mind map of SOA themes and their categories along with the relevant topics and research methodologies used in each theme. As presented in Fig. 9, in the SOA Practice, the majority of the articles used a qualitative methodology with 23

Table A.1

ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S1	Service oriented computing in practice – an agenda for research into the factors influencing the organizational adoption of service oriented architectures	Luthria & Rabhi (2009)	This review explored the state of technology and examined the significant factors of adopting service-oriented computing.	–	Review	Review	–	–	–	Others	SOA Adoption
S2	Factors influencing the alignment of SOA development with business objectives	Antikainen & Pekkola (2009)	This paper explored a better understanding of SOA success factors from the perspective of business and IT alignment.	–	Multiple case study	Qualitative	Interview	IT and Business experts from 9 companies	Finland	AISel	SOA Impact
S3	An Exploratory Study of Risk Factors for Implementing Service-Oriented IS Projects	Hsin-Lu Chang & Chia-Pei Lue (2009)	This study investigated SOA risk factors of Information Systems projects and examining the influence of these factors.	IBM Service Integrated Maturity Model (SIMM)	Survey	Quantitative	Questionnaire	105 IS department staff	Taiwan	Springer	SOA Adoption
S4	Adopting SOA – Experiences from nine Finnish organizations	Kokko et al. (2009)	This study investigated the obstacle and challenges of adopting SOA in large Finnish organisations.	–	Multiple case study	Qualitative	Interview	12 IT experts and managers of nine Finnish companies	Finland	IEEE	SOA Adoption
S5	Building the Business Case for SOA: A Study of the Business Drivers for Technology Infrastructure Supporting Financial Service Institutions	Haresh Luthria & Fethi A. Rabhi (2009)	This research examined the utilisation of SOA in fifteen companies to find the business drivers for adopting SOA.	–	Multiple case study	Qualitative	Interview	15 IT experts and top managers from 13 companies	Multi-country	Springer	SOA Adoption
S6	TinySOA: a service-oriented architecture for wireless sensor networks	Lopez and Blobel [26]	This paper proposed a service-oriented architecture, namely TinySOA, to help programmers for accessing wireless sensor networks through the applications.	TAM	Experiment	Mixed method	Observation-Questionnaire	13 programmers	–	Springer	SOA Practice
S7	Enablers and Benefits of Implementing Service-Oriented Architecture: An Empirical Investigation	Wai Fong Boh & Daniel Yellin (2010)	This paper investigated the key factors and benefits of SOA implementation in a group of early adopter organisations.	Conceptual model	Survey	Quantitative	Questionnaire	90 IT architectures (IBM clientsOpen group members, and Conference participants)	–	Others	SOA Adoption
S8	Critical Success Factors for the Implementation of ICT Projects	Klara Antlova (2010)	This article identified the CSFs that affect the adoption of ICT projects with a SOA perspective.	Conceptual model	Multiple case study	Mixed method	Questionnaire, Interview, Literature Review	Managers and the company owners from 50 Czech Republic companies	Czech Republic	Springer	SOA Adoption
S9	Critical Success Factors in SOA Implementation: An Exploratory Study	Jung Hoon Lee, Hye-Jung Shim, & Kyung Kyu Kim (2010)	This article identified success factors of SOA adoption based on review of 34 papers and 22 interviews.	–	Multiple case study	Qualitative	Interview, Literature Review	22 developers, managers, and senior managers from 12 medium-to-large Korean companies	Korea	Taylor & Francis	SOA Adoption

(continued on next page)

Table A.1 (continued).

ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S10	The Potential of Service-Oriented Architectures	Gerić (2010)	This study investigated the benefits of SOA implementation in Croatia.	–	Survey	Quantitative	Questionnaire	34 experts from Croatian ICT companies.	Croatia	IEEE	SOA Adoption
S11	Adopting and Evaluating Service Oriented Architecture in Industry	Nasr et al. (2010)	This study presented a descriptive single case study by reengineering and further evolution of adopting SOA in Transport and logistics domain.	–	Single case study	Qualitative	Observation	–	–	IEEE	SOA Adoption
S12	Readiness and Maturity of Service-oriented Architectures in the German Banking Industry	Eckert et al. (2010)	This study identified the important issues regarding SOA readiness and maturity through a case study in a German bank.	SOA Maturity Model	Single case study	Qualitative	Interview	4 IT architecture of German banks	Germany	IEEE	SOA Practice
S13	A Literature Review of Research on Service Oriented Architectures (SOA): Characteristics, Adoption Determinants, Governance Mechanisms, and Business Impact	Joachim (2011)	This review synthesised the existing literature about SOA from the perspective of business.	–	Review	Review	Literature Review	–	–	AISel	SOA Impact
S14	Service Oriented Architecture Adoption Management Roadmap	Moeini et al. (2011)	This article presented a roadmap for migrating to SOA by synthesizing the existed roadmaps.	–	Review	Review	–	–	–	IEEE	SOA Adoption
S15	Critical Success Factors of Service Orientation in Information Systems Engineering	Aier et al. (2011)	This article identified the CSFs of SOA implementation in IS engineering, develop and evaluate a conceptual model based on a survey.	Conceptual model	Survey	Quantitative	Questionnaire	289 IT professionals	Switzerland, Germany, and Austria	Springer	SOA Adoption
S16	Main Factors Affecting the Adoption and Diffusion of Web Service Technology Standards	Hu [47]	This review paper introduced the technology of web services (WS) standards based on TOE framework and Diffusion of Innovation and analysis the key factors influencing WS technology standards.	DOI and TOE	Review	Review	–	–	–	Springer	SOA Concept
S17	SOA Migration Case Studies and Lessons Learned	Bhallamudi & Tilley (2011)	This paper examined how the case studies of this research successfully migrate to SOA and to conclude the lessons learned from these organisations.	Evolution Process Framework for SOA	Multiple case study	Qualitative	Secondary data	Three case studies and a report	USA	IEEE	SOA Practice
S18	SOA Adoption Phases: A Case Study	Trkman et al. (2011)	This research investigated the main risks of implementing SOA.	–	Single case study	Qualitative	Interview, internal documents, and procedures.	Senior executives, directors, and employees	Slovenia	Springer	SOA Practice

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S19	Enterprise SOA Implementation Readiness: A Case Study in Malaysia	Abdul Manan & Hyland (2011)	This research developed and validate an Enterprise SOA readiness framework suitable for public sector organisations.	–	Multiple case study	Quantitative	Structured Interview	5 ICT consultancy in one public sector agency and 7 senior IT practitioners in another public sector agency	Malaysia	AISEL	SOA Adoption
S20	Realizing service migration in industry—lessons learned	Nasr et al. (2011)	This article investigated the advantage and disadvantage of implementing SOA in large enterprises to find out the actual benefits of SOA in practice.	–	Multiple case study	Qualitative	Observation, Interview	Project manager, business architect, business analysts, SOA architects, and developers	–	Wiley Online Library	SOA Practice
S21	Migration of information systems in the Italian industry: A state of the practice survey	Torchiano et al. (2011)	This paper investigated the state of the practice of software migration (such as SOA) among Italian companies.	–	Survey	Quantitative	Questionnaire (Structured interview or open-ended questionnaire)	59 project managers and IT managers, developers, and IT architects from 59 Information Technology companies and IT departments	Italy	Science Direct	SOA Practice
S22	Assessing IT-business alignment in service-oriented Enterprises	Chang et al. (2011)	This article proposed a framework regarding IT and business alignment to improve the quality of alignment in Service-oriented based organisations.	IBM's Service Integration Mutuality Model (IBM SIMM)	Survey	Quantitative	Questionnaire	96 employees of companies that implemented service integration concepts for their customer service systems.	–	AISEL	SOA Impact
S23	Does SOA create or require IT/business collaboration? investigating SOA's potential to reduce the gap between IT and business	Joachim et al. (2011)	This study identified the benefits of SOA and investigated the value of SOA in organisations.	Conceptual model	Survey	Quantitative	Questionnaire	122 IT architecture managers	Germany	AISEL	SOA Impact
S24	The joint impact of service-oriented architectures and business process management on business process quality: an empirical evaluation and comparison	Daniel Beimbom & Nils Joachim (2011)	This research examined the effect of SOA and BMP on business process quality dimension including straight through processing, standardisation and consolidation, B2B integration, and quality control.	Conceptual model	Survey	Quantitative	Questionnaire	157 IT architect managers	Germany	Springer	SOA Impact
S25	Service Oriented Architecture Adoption Trends: A Critical Survey	Seth et al. (2012)	This review paper investigated the related factors of SOA implementation and to identify the level of their importance for implementing SOA.	–	Review	Review	–	–	–	Springer	SOA Adoption

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S26	Success Factors Model: Case Studies in the Migration of Legacy Systems to Service Oriented Architecture	Galiniun & Shahbaz (2012)	This research explored the effective and success factors of SOA migration from the technical and business aspect in five companies.	–	Multiple case study	Qualitative	Interview	Business and technical members from 5 European companies	Multi-country	IEEE	SOA adoption
S27	An exploratory survey on SOA knowledge, adoption and trend in the Italian industry	Maurizio Leotta et al. (2012)	This research explored the level of SOA knowledge, the level of SOA adoption and the intention rate of implementing SOA among Italian companies.	–	Survey	Quantitative	Structured Interview (Structured interview or open-ended questionnaire)	159 project managers, architects, developers	Italy	IEEE	SOA Adoption
S28	Promises from SOA: Reengineering a procurement process at Belgacom Mobile—a case study approach	Joachim Van den Bergh & Stijn Viaene (2012)	This paper supported the application of a combined Business Process Management and SOA approach and contributed to the body of knowledge on the business value of SOA.	–	Single case study	Qualitative	Observation	Supply chain manager and the external consultants of Belgacom Mobile (Today is known as Proximus Company)	Belgium	Emerald Insight	SOA Impact
S29	Service-Oriented Architecture Adoption: A Normative Decision Model	Ciganek et al. (2012)	This paper developed a normative decision making model for defining significant factors which affect timing and approach of SOA adoption based on analyzing multiple case studies.	A normative decision making model	Multiple case study	Qualitative	Interview	17 SOA experts and IS Managers from 8 major national organisations	USA	IGI Global	SOA Practice
S30	A Proposition of Critical Success Factors Influencing SOA Implementation in Healthcare	Koumaditis et al. (2012)	This research investigated CSFs related to SOA implementation in healthcare organisations.	Critical success factor model	Single case study	Qualitative	documentation, interviews, direct observation, participant observation and archival records	4 professionals from a Public Insurance Organisation	–	AISel	SOA Adoption
S31	A Framework for Assessing Enterprise-Wide SOA Implementation Readiness	Mahshitah Abdul-Manan & Peter Hyland (2013)	This article developed a critical success factor framework and a related scorecard to evaluate the readiness of an organisation in order to implement Enterprise-SOA (ESOA).	Critical success factor model	Single case study	Design science	Structured Interview	12 project leaders and managers	Malaysia	IGI Global	SOA Adoption
S32	The Identification of Service Oriented Architecture-Specific Critical Success Factors	Ian Owens & John Cunningham (2013)	This research identified the CSFs of implementing SOA based on reviewing literature and conducting a survey on a large organisation.	–	Survey	Quantitative	Questionnaire	55 project managers and implementers	–	Others	SOA Adoption
S33	SOA implementation critical success factors in healthcare	Koumaditis et al. (2013)	This work identified CSFs for implementing SOA in healthcare industry through literature review and a case study.	Conceptual model	Single case study	Qualitative	Documentation, semi-structured interviews, observation, archival records	Five professionals (2 top-managers, 2 project managers and 1 administrator)	–	Emerald Insight	SOA Adoption

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S34	SOA adoption in e-banking	Basias et al. (2013)	This article investigated and evaluated the factors influencing SOA adoption in E-banking and developed a conceptual model to describe the factors.	Conceptual model	Single case study	Qualitative	Interview, archival records, documentation and observation	Four key role experts in SOA related project in a European bank	–	Emerald Insight	SOA Adoption
S35	Factors Effecting Service Oriented Architecture Implementation	Waris et al. (2013)	This review paper focused on effective factors of designing and implementing SOA to use as a guideline for SOA projects team leaders.	Critical success factor model	Review	Review	–	–	–	IEEE	SOA Adoption
S36	Implementing a Service-Oriented Architecture: A Technochange Approach	Eli Hustad & Lene Staverlökk (2013)	This study found the organisational challenges and system development issues regarding implementing SOA.	BEA SOA domain model	Action Research	Qualitative	Semi-structured interviews, observation, and secondary data (project documents, IT portfolio management reports, and institutional rules and documents)	15 top management, line managers, project leaders, system developers, and test engineers.	Norway	Springer	SOA Practice
S37	The Impact of SOA Implementation on IT-Business Alignment: A System Dynamics Approach	Jae Choi, Derek L. Nazareth & Hemant K. Jain (2013)	This study developed a dynamic model for examining the impact of SOA implementation strategies in different external business environment and organisational context on the business and Information Technology alignment and Information Systems cost.	SOA-enabled IT-business alignment model	Experiment	Design science	Simulation	–	–	ACM DL	SOA Impact
S38	The influence of SOA governance mechanisms on IT flexibility and service reuse	Joachim et al. (2013)	This paper empirically identified the most useful mechanisms of SOA governance for achieving the SOA benefits.	–	Survey	Quantitative	Questionnaire	81 IT architecture managers from SOA-using firms	Germany	Science Direct	SOA Impact
S39	Implementing Service Oriented Architecture – a Case Study	Ashish Tewary & Parag Kosalge (2013)	This exploratory paper investigated the issues and procedures of implementing a pilot SOA project in an oil company to understand the challenges involving SOA adoption in practice.	–	Single case study	Qualitative	Interview and Observation	SOA project consultants and the company employees	–	Inder-science	SOA Practice
S40	Factors affecting the organizational adoption of service-oriented architecture (SOA)	MacLennan and Van Belle [48]	This article studied the organisational adoption of SOA in South Africa by integrating TOE framework and DOI theory based on a survey.	TOE and DOI	Survey	Quantitative	Questionnaire	111 IT experts	South Africa	Springer	SOA Adoption

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S41	Service-oriented Architecture Projects in Practice: A Study of a Shared Document Service Implementation	Eli Hustada, Christer de Lange (2014)	This study developed a share document service in practice based on SOA technology to support different business sectors by focusing on a governmental organisation.	–	Single case study	Qualitative	Interview	15 initial interviews, semi-structured with system developers, system architects, and project managers	Norway	Science Direct	SOA Practice
S42	Towards an Integrated SOA-based Architecture for Interoperable and Responsive Manufacturing Systems Using the ISA-95 Object Model	A. García-Domínguez et al. (2014)	Three main problems of manufacturing systems have solved by the proposed Holonic manufacturing based on SOA and a Multi-Agent System.	–	Multiple case study	Design science	–	–	Spain	Others	SOA Practice
S43	Design and value of service-oriented technologies for smart business networking	Rainer Alt & Martin Smits (2014)	This study identified the business value of smart business networking based on SOA.	Conceptual model	Single case study	Qualitative	Interview, Secondary Data	7 representatives from a centralized bank	Germany	AISel	SOA Impact
S44	A Framework for Service-oriented Architecture Adoption in e-Banking: the Case of Banks from a Transition and a Developed Economy	Marinos Themistocleous Nikolaos Basias & Vincenzo Morabito (2015)	This research proposed an SOA adoption methodological framework, evaluated the framework based on two case studies and compared the findings of the case studies.	Conceptual model	Multiple case study	Qualitative	Interviews, documentation, and observation	One project manager of SOA adoption, one IT manager, one business manager, and one IT strategy member	Greece and Bulgaria	Taylor & Francis	SOA adoption
S45	Organizational structures during SOA implementation: the case of a Greek healthcare organization	Koumaditis & Marinos Themistocleous (2015)	This research identified organisational studies related to SOA in Healthcare, and proposed and evaluated the findings in a public case study.	–	Single case study	Qualitative	Documentation, semi-structured interviews, direct observation, participant observation and archival records	2 top-managers, one internal project manager, one external project manager, and one administrator from a public healthcare organisation	Greece	Emerald Insight	SOA Practice
S46	A Service-Oriented Architecture Implementation in the Digital Factory of the University	Jeffrey Wermann et al. (2015)	This paper presented a summary of implementing SOA concept in a digital factory of the University of Emden.	–	Experiment	Qualitative	Simulation	A simulated digital factory at the University of Emden/Leer	Germany	Springer	SOA Practice
S47	Service-Oriented Architecture as a Driver of Dynamic Capabilities for Achieving Organizational Agility	Hareesh Luthria & Fethi A. Rabhi (2015)	This article proposed a conceptual framework for analysing and testing the basic characteristics of an SOA compatible architecture to attain the dynamic capabilities, namely: integration of internal resources, integration of external resources, rapid product development, learning, and creation of assets.	Conceptual model	Multiple case study	Qualitative	Interview	Developers in 15 firms	–	Springer	SOA Practice

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S48	Proposing and Testing SOA Governance Process: A Case Study Approach	Koumaditis & Marinos Themistocleous (2015)	This work developed SOA Governance Process (GP) based on SOA Governance framework to precise guidelines and attributes of SOA GP and to attain the appropriate degree of SOA governance successfully.	SOA Governance framework	Single case study	Qualitative	Interview	Top managers	Greece	IGI Global	SOA Practice
S49	Understanding the Dynamics of Service-Oriented Architecture Implementation	Xitong Li & Stuart E. Madnick (2015)	This study developed a system dynamics (SD) model for SOA implementation by adopting a process perspective and using a simulation analysis to investigate the phenomenon of “tipping point”.	System Dynamics Model	Multiple case study	Qualitative	Semi-structured Interview and Literature review	10 interviews with 14 IT managers and developers – 15 interviews with 22 interviewees including CIOs, mid-level IT managers, and IT developers	USA	Taylor & Francis	SOA Practice
S50	Achieving Business Agility Through Service Oriented Architecture in Recovering Markets	Bharadwaj et al. (2015)	This paper designed a conceptual model to handle the agility of business value chain by implementing SOA in organisations.	Conceptual model	Not clear	Conceptual Study	–	–	–	Springer	SOA Impact
S51	Governance of Service-Oriented Architecture in a Healthcare Organization: A Case Study	Koumaditis (2015)	This research developed and examined an SOA governance framework, as a decision supporting tool, to successfully implement SOA in healthcare industry through a large public hospital.	SOA governance framework	Single case study	Qualitative	Documentation, interviews, direct observation, participant observation, and archival records,	Two top-managers, an internal project manager, an external project manager, and an administrator.	–	Taylor & Francis	SOA Practice
S52	SOA Strategic Vision: Role in Establishing Strong SOA Governance	George Joukhadar & Fethi Rabhi (2015)	This study improved the understanding of SOA governance and investigated the role of SOA vision in implementing strong SOA governance in different companies.	–	Not clear	Qualitative	Interview	28 SOA/IT governance experts	Multi-country	AISeL	SOA Concept
S53	Adaptive business network systems: A service-oriented architectural approach	He et al. (2015)	This research paper proposed an adaptive business network systems (ABNS) to support the collaboration over organisational boundaries and modified the business strategies in a comprehensive framework to implement an adaptive network.	Adaptive Business Network Systems Framework	Not clear	Conceptual Study	–	–	–	IEEE	SOA Concept
S54	Impact of Service-Oriented Architecture adoption in information system	Fernando et al. (2015)	This paper explored the effect of implementing SOA based on the literature review.	–	Review	Review	–	–	–	IEEE	SOA Impact

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S55	Towards the next generation service oriented enterprise architecture	Rabelo et al. (2015)	This study presented how Enterprise Architecture is an essential key enabler of SOEA through EA approaches.	Generalized Enterprise Reference Architecture modelling framework	Review	Review	–	–	–	IEEE	SOA Impact
S56	Proposing a new framework for service oriented architecture adoption	Niknejad et al. [49]	This study proposed an integrated framework based on TOE and Hot-fit model to investigate the most important factors in motivating decision makers to adopt SOA in organisations.	TOE and Hot-fit model	Review	Conceptual Study	–	–	–	IEEE	SOA Adoption
S57	SOA in practice – a study of governance aspects	Joukhadar & Rabhi (2016)	This research investigated which aspects of SOA should be considered by organisations for adopting SOA governance framework.	–	–	Qualitative	Interview	28 IT/SOA governance professionals	Multi-country	Springer	SOA Practice
S58	Anatomy of one service-oriented architecture implementation and reasons behind low service reuse	Alkkiomäki & Smolander (2016)	This research examined the factors and barriers that cause the low ratio of service reusability through exploring an SOA implemented case study in manufacturing industry.	–	Single case study	Qualitative	In-depth interviews	Two integration architects from a manufacturing company	–	Springer	SOA Practice
S59	The Role of Service-Oriented Architecture as a Part of the Business Model	Alkkiomäki (2016)	This article used an ontological approach as a tool to make SOA a merging part of the business model and to identify the conceptual link of IT and business value in organisations.	Osterwalder business model ontology	Single case study	Qualitative	Interview	IT and business representatives (preliminary information)	–	Inder-science	SOA Impact
S60	In Search of Explanations: Conceptualizing the Relationship between Service-oriented Architecture and Organizational Agility	Janeke Richter & Dirk Basten (2016)	This paper conceptualised the relationship between service-oriented architecture and Organisational Agility (OA).	–	Review	Review	–	–	–	AISel	SOA Concept
S61	Identifying best practices in organisational SOA governance adoption: case study of Saudi Arabia's e-government programme	Bader Alghamdi, Leigh Ellen Potter, & Steve Drew (2016)	This work investigated the best key practices of successful adoption of SOA governance based on the CSF framework through a case study (Saudi Arabia e-government programme).	Critical success factor model	Single case study	Qualitative	Semi-structured interviews	Five project manager, senior enterprise architect (EA), government service bus (GSB) manager, business analyst and integration specialist	Saudi Arabia	AISel	SOA Practice
S62	Adoption of emerging technologies into ERP systems landscape: A South African study	Jean-Jacques Du Plessis & Gwamaka Mwalemba (2016)	This study identified the influential factors for adopting the emerging technologies such as SOA and CC in the enterprise resource planning (ERP) of South Africa.	–	Multiple case study	Qualitative	Semi-structured Interview	7 IT experts/decision makers from seven organisations	South Africa	IEEE	SOA Adoption

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S63	Empirical insights into the development of a service-oriented enterprise architecture	Alwadain et al. (2016)	This research analysed the evolution process of Enterprise Architecture based on a government case study that has introduced SOA as an influential technology on the evolution of EA.	Archer's morphogenetic theory	Single case study	Qualitative	Interview, Secondary Data	8 senior executives	United Arab Emirates	Science Direct	SOA Practice
S64	Trust-based Service-Oriented Architecture	Aljazzaf et al. (2016)	This work proposed a trust based SOA solution based on Total Trust Evaluator Framework (ToTEF) to improve the trust among different parties of SOA.	Trust-based SOA based on ToTEF (SOA extended based on a trust framework)	Experiment	Quantitative	Simulation	–	–	Science Direct	SOA Practice
S65	SOA maturity model a frame of reference	Supriya Pulparambil & Youcef Baghdadi (2016)	This review paper proposed the main aspects of maturity models such as scope, reference, process execution, perspective maturity, and governance for accelerating SOA adoption in organisations.	SOA Perspective Maturity Assessment	Review	Review	–	–	–	IEEE	SOA Adoption
S66	Implementation of a city-wide Health Information Exchange solution in the largest metropolitan region in China	Li Zhou, David W Bates, Jianlei Gu, & Hui Lu (2016)	This paper explained the implementation of Health Information Exchange (HIE) system based on SOA in Shanghai, China.	–	Single case study	Qualitative	Observation	–	China	IEEE	SOA Practice
S67	A service-oriented approach for integrating broadcast facilities	Alexis Huf, Ivan Salvadori, & Frank Siqueira (2016)	This paper proposed the adoption of an SOA for controlling broadcasting equipment, addressing difficulties specific to this environment such as the prevalence of non-Web Services, embedded devices, and constrained computational resources.	–	Experiment	Quantitative	Simulation	–	–	IEEE	SOA Practice
S68	Service-Oriented Architectures for Interoperability in Industrial Enterprises	Ahmed Ismail & Wolfgang Kastner (2017)	This book chapter examined the technology choice and architecture design of five main projects of Service-Oriented Reference Architecture (SORA) and to present some features and limitations of these projects.	–	Review	Review	–	–	European Union	Springer	SOA Concept
S69	An event-driven manufacturing information system architecture for Industry 4.0	Theorin et al. (2017)	This paper designed an event based SOA, called Line Information Systems Architecture (LISA), to increase the flexibility of integrating manufactories and the utilisation of data.	–	Single case study	Design science	–	–	–	Taylor & Francis	SOA Practice

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S70	A methodology to identify health and social care web services on the basis of case stories	Pecoraro et al. (2017)	This project proposed a comprehensive methodology with the ability of discovering all practical requirements and web services through analysing case stories related to healthcare and their involving stakeholders.	–	–	Qualitative	Semi-structured interviews	Patient, General Practitioner, specialist, pharmacist	Italy	IEEE	SOA Practice
S71	Quantitative analysis of information leakage in service-oriented architecture-based Web services	Kushal Anjaria & Arun Mishra (2017)	This study quantitatively investigated the dynamic information process of an SOA-based hypothetical case study scenario.	A threat model	Single case study	Quantitative	Hypothetical scenario	–	–	Emerald Insight	SOA Practice
S72	Exploring the main building blocks of SOA method: SOA maturity model perspective	Pulparambil et al. (2017)	This study proposed a conceptual framework for explaining the building blocks of SOA techniques at different stage of evolution and validated the proposed model of the study through a case study.	SOA maturity models (SOAMMs)	Single case study	Qualitative	Interview and Questionnaire	SOA delivery team of a university	Oman	Springer	SOA Practice
S73	Service Oriented Architecture adoption model for ICT office in Malaysia	Abbas et al. (2017)	This effort identified the crucial factors of SOA adoption and developed a TOE framework for improving the decision making process in Malaysian organisations.	TOE	Survey	Quantitative	Questionnaire	289 IT and SOA experts from 5 SOA adopted companies	Malaysia	IEEE	SOA Adoption
S74	Adopting SOA in Public Service Provision	Mesfin et al. (2017)	This study identified the e-readiness for principles and technologies of SOA applications in implementing ubiquitous public service in Ethiopia.	e-Readiness Assessment Model	Archival Research	Qualitative	Secondary data (government documents, strategic documents etc.)	–	Ethiopia	Springer	SOA Adoption
S75	Interoperability Framework for Government E-services of the Philippines	Regalado Reyes Jr. & Marivic Tangkeko (2017)	This study investigated how the e-services of government can be unified by applying the whole-of government strategy.	Conceptual model	Multiple case study	Qualitative	Interview (focus group discussion) Secondary data (document analysis and website visits)	IT team of four government agencies	Philippines	AISel	SOA Practice
S76	Explaining The Role of Service-Oriented Architecture for Cyber-Physical Systems by Establishing Logical Links	Gruettner et al. (2017)	This paper explored the role of SOA for implementing cyber-physical systems through a concept-driven review.	–	Review	Review	–	–	–	AISel	SOA Concept

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S77	Framework for the development of Business intelligence using computational intelligence and service-oriented architecture	Yamid Fabián Hernández-Julio et al. (2017)	This article tried to merge the computational intelligence techniques in business intelligence systems by developing an SOA based framework to assist collecting pre-information and its diffusion for helping managers in making decision procedure.	Computational intelligence-based framework	Single case study	Qualitative	Observation	Observation of physiological variables on Holstein cows in an agricultural enterprise	–	IEEE	SOA Practice
S78	Constructing service-oriented architecture adoption maturity matrix using Kano model	Mohd Hamdi Irwan Hamzah et al. (2017)	This paper developed a technique based on Kano Model which can be used for creating a matrix to examine and evaluate the adoption of SOA considering Business and IT benefits.	Kano Model	Not clear	Quantitative	Questionnaire	–	–	Others	SOA Impact
S79	OSaaS: Online shopping as a service to escalate E-commerce in developing countries	Maqbool Khan et al. (2017)	This paper disclosed the obstacles of Electronic Commerce in developing countries through providing a model of Online Shopping as a Service based on CC and SOA.	Online Shopping as a Service (OSaaS) Transaction Framework	Not clear	Quantitative	Secondary Data	ITU World Telecommunication data from 2005–2015 for developed and developing countries.	–	Others	SOA Impact
S80	Advances in internet of things (IoT) in manufacturing	Rakshith Badarinath, & Vittaldas V. Prabhu (2017)	This study proposed a model based on SOA for understanding the application of IoT in manufacturing industry such as predictive maintenance, shop floor automation, smart workers, and energy aware manufacturing.	SOA Model for IoT applications	Review	Review	–	–	–	Springer	SOA Practice
S81	ArchSORS: A Software Process for Designing Software Architectures of Service-Oriented Robotic Systems	Oliveira et al. (2017)	This research proposed a new process, named Architectural Design of Service-Oriented Robotic System (ArchSORS) to support the design of a Service-Oriented Robotic System (SORS) software architectures.	–	Experiment	Quantitative	Observation	30 students of a French University	France	Others	SOA Practice
S82	Patient Monitoring System Based on e-health Sensors and Web Services	Hameed et al. (2017)	This article developed a patient monitoring system by applying SOA and some medical sensors.	Conceptual patient monitoring system framework	Experiment	Design science	Simulation	60 healthy male and female applicants	–	IEEE	SOA Practice

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ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S83	A Construction of Service-Oriented Architecture Adoption Maturity Levels using Adoption of Innovation Concept and CMMI	Hamzah et al. (2018)	This study constructed the SOA adoption maturity level by integrating the concept of innovation adoption and Capability Maturity Model Integration.	SOA Maturity Model	Review	Conceptual Study	–	–	Malaysia	Other	SOA Adoption
S84	Service Oriented Architecture (SOA) implementation: Success factors and realized benefits	Tom Yoon & Bong-Keun Jeong (2018)	This study found the success factors of SOA implementation through multiple case studies and validated the findings by conducting a survey among SOA professionals.	–	Multiple case study	Mixed method	Press releases, annual reports, interviews with top management, conference presentations, & questionnaire	Top managements of 12 organisations and SOA professionals	USA	IGI Global	SOA Impact
S85	Every Second Counts: Integrating Edge Computing and Service Oriented Architecture for Automatic Emergency Management	Lei Chen and Cristofer Englund (2018)	This research provided an SOA-based collaboration framework for management of emergency events that regulate emergency response systems with stakeholders and completely automatise the rescue method.	–	Experiment	Design science	Simulation	–	Sweden	Others	SOA Impact
S86	Investigating the Impact of System Quality on Service-Oriented Business Intelligence Architecture	Lee-Kwun Chan and Phooi-Yee Lau (2018)	This quantitative research examined the impact of system quality on the SOA-based business intelligence architecture according to the information systems success model.	Delone and Mclean's Information System success model	Survey	Quantitative	Questionnaire	60 IT practitioners	Malaysia	SAGE	SOA Impact
S87	Limiting Technical Debt with Maintainability Assurance – An Industry Survey on Used Techniques and Differences with Service- and Microservice-Based Systems	Justus Bogner et al. (2018)	This research explored the industry state of practice of SOA-based and microservice-based systems through distributing a questionnaire among software professionals.	–	Survey	Quantitative	Questionnaire	60 software professionals	Germany	IEEE	SOA Practice
S88	State of the practice in service identification for SOA migration in industry	Abdellatif et al. (2018)	This research explored the state of SOA migration in the industry to realise the migration process and service identification by surveying 45 experts.	–	Survey	Mixed method	Questionnaire and semi-structured interview	45 industrial experts of legacy-to-SOA migration	Multi-country	Springer	SOA Practice
S89	Web Service Composition on Smartphones: The Challenges and a Survey of Solutions	Mesfin et al. (2018)	This research conducted a literature review for exploring the existing Web services and the relevant technologies to create and design Web services in the domain of smartphones.	–	Review	Review	Literature Review	–	–	Springer	SOA Concept

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Table A.1 (continued).

ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S90	SOA maturity influence on digital banking transformation	Megargel et al. (2018)	This explanatory research examined the relationships of the technology infrastructure, SOA maturity, IT governance processes, and their influence on time-to-market in digital banking services and products.	SOA maturity model	Multiple case study	Mixed method	Interview and Observation	Chief technologist of 8 banks	Asian countries	Others	SOA Impact
S91	SOA Adoption Factors in E-Banking: An Empirical Analysis from the Practical Perspective	Siadat et al. (2019)	This study identified and analysed the significant factors of SOA adoption in banking sector.	Conceptual model	Survey	Quantitative	Questionnaire	70 bank employees	Iran	IGI Global	SOA Adoption
S92	Directions for future research on the integration of SOA, BPM, and BRM	Fischer et al. (2019)	This research explored the present state of investigation on the holistic combination of SOA, BRM, and BPM domain and studied the pairwise integration of these concepts in scenarios.	–	Review	Review	Literature Review	–	–	Emerald Insight	SOA Practice
S93	An exploratory study for investigating the issues and current practices of Service-Oriented Architecture adoption	Hamzah et al. [1]	This research examined practices and issues of SOA adoption, the significance of business and IT benefits of SOA adoption.	–	Grounded Theory	Qualitative	Interview	14 SOA practitioners	Malaysia	Universiti Utara Malaysia Press	SOA Adoption
S94	A research on SOA in the IT industry of Pakistan	Momina Shaheen et al. [2]	This study analysed the SOA usage among IT companies in Pakistan.	–	Survey	Qualitative	Questionnaire and semi-structured interview	SOA professionals	Pakistan	ACM DL	SOA Adoption
S95	A decision support system for healthcare system quality improvement in blood centres: a case from Turkey	Burak Erkeyman & M. Sefa Gör (2019)	This study proposed an SOA-based system for improving the efficiency, reliability, and speed of blood centres.	–	Single case study	Design science	Data extracted from the system	–	Turkey	Taylor & Francis	SOA Practice
S96	Development, implementation and preliminary results of an electronic reminder for HIV screening using a service oriented architecture	Rubin et al. (2019)	This study explored the development and implementation of an SOA-based electronic reminder for human immunodeficiency viruses (HIV) screening for interacting from the electronic health record with the Fast Healthcare Interoperability Resources server and Clinical Decision Support service triggered by medical professionals' workflow in the electronic health record.	–	Experiment	Quantitative	Data extracted from the system	–	Argentina	IOS Press	SOA Practice

(continued on next page)

Table A.1 (continued).

ID	Title	Author/Year	Objective	Theory/Model	Research strategy	Methodology	Data collection methods	Subject	Country	Online database/publisher	Theme
S97	E-Marketing Of New Student Admissions Through A Service Oriented Architecture Approach	Muhamad Muslih (2019)	This research developed and evaluated a new SOA-based e-marketing system for new students' admission in Indonesian universities.	–	Survey	Quantitative	Questionnaire	52 high schools students	Indonesia	Others	SOA Practice
S98	Guiding the selection of service identification methods based on SOA maturity level	Basel Bani-Ismael and Youcef Baghdadi (2019)	This paper proposed a novel selection approach for service identification methods (SIMs) that employees the desired SOAMM as guidance to assist the enterprises in determining an appropriate SIM among the present evaluation frameworks.	SOA maturity level	Not clear	Quantitative	–	–	Oman	Emerald Insight	SOA Practice
S99	SOAP-Based Web Service for Localization of Multi-robot System in Cloud	Tlijani Hayet and Jilani Knani (2019)	This study developed and evaluated a software solution for mobile robot localisation using CC and SOA.	–	Experiment	Quantitative	Simulation	–	Tunisia	Springer	SOA Practice
S100	Cloud computing as a facilitator for web service composition in factory automation	Juha Puttonen et al. (2019)	This study proposed and examined a new approach that used the resources of CC in the automatic composition of SOA-based web services.	Web ServiceComposition Framework	Experiment	Quantitative	Simulation	–	–	Springer	SOA Practice
S101	SOA Issues and their Solutions through Knowledge Based Techniques-A Review	Abbas et al. (2019)	This paper explored some important issues confronted by the SOA community such as service quality and service selection through knowledge management techniques.	–	Review	Review	Literature Review	–	–	Others	SOA Concept
S102	Service-Oriented Architecture Application in Long-Term Care Institution: A Case Study on an Information System Project Based on the Whole Person Concept in Taiwan	Feng-I Chung & Chia Lun Lo (2019)	This investigation developed and examined a new system and integrated different operating systems based on the SOA concept, called the UCare system, a ubiquitous care system for aiding long-term care in hospitals.	DeLone and McLean IS Success Model	Single case study	Quantitative	Questionnaire	20 staffs	Taiwan	Others	SOA Practice
S103	Service-oriented architecture of environmental information systems to forecast the impacts of natural disasters in South Korea	Euisung Jung & Eun Ju Jung (2019)	This research examined the environmental information systems (EISs) and proposed an SOA-based system for improving the quality of EISs to anticipate the consequences of natural disasters in South Korea.	DeLone and McLean's IS success model	Survey	Design science	Questionnaire	38 system users	South Korea	Emerald Insight	SOA Practice

papers. In addition, 13 studies applied the qualitative methodology under SOA Adoption followed by 4 studies under SOA Impact and one paper under SOA Concept. It is worth mentioning that most of these qualitative studies applied a case study approach as a main research strategy. Furthermore, out of the 29 quantitative studies, 10 papers took up the SOA Adoption theme while 7 studies were related to SOA Impact. The remainder of the quantitative papers targeted SOA Practice (12 papers). The majority of these quantitative studies collected data through surveys. On the other hand, a few articles used the design science methodology, with 5 studies falling under SOA Practice, two papers under SOA Impact, and one article under SOA Adoption. Furthermore, the mind map shows four conceptual studies surrounding some of these themes, with two studies in SOA Adoption and one study in each group taking on the SOA Concept and SOA Impact. In addition, 15 studies were review papers, 6 studies took on SOA Concept, 4 studies on SOA Adoption, 3 studies on SOA Impact, and two studies on SOA Practice. Finally, three studies used the mixed method, with two studies targeting SOA Practice and one taking on SOA Adoption.

RQ (2): What theoretical frameworks and models are emphasised in current SOA research?

Out of the 103 primary studies, 50 papers did not apply any model or theory while 13 articles proposed unique conceptual models based on their study findings. Because of the organisational nature of SOA research, all the remaining primary studies employed organisational theories and models. Based on the findings presented in this section, a few studies had applied an IS adoption theories and frameworks such as the Technology-Organisation-Environment (TOE) framework, the Technology Acceptance Model (TAM), and the Diffusion of Innovation (DOI) theory, while the majority of the articles used SOA theories developed in a non-IS field. For example, Hu [47] used the TOE framework and the DOI theory for categorising and examining the factors affecting the adoption of Web Services. Similarly, MacLennan and Van Belle [48] integrated the TOE framework and the DOI theory by conducting a survey to analyse the organisational adoption of SOA in South Africa. In a conceptual study, Niknejad et al. [49] proposed an integrated model based on the TOE framework and the Human-Organisation-Technology fit (Hot-fit) model to identify the most significant factors that motivated decision-makers to adopt SOA in enterprises. Elsewhere, Avilés-López and García-Macías [50] applied the Technology Acceptance Model (TAM) and a quantitative methodology to evaluate the model proposed in their study. Fig. 10 illustrates the theoretical models and frameworks used in the primary studies.

RQ (3): What are the most influential factors affecting SOA adoption/implementation in organisations?

To answer RQ3, all significant factors were extracted from the primary studies. Fig. 11 presents the most influential factors that have been mentioned more than 2 times in the primary studies (41 factors). As shown in Fig. 11, the most controversial factor among researchers was SOA Governance, with 30 studies mentioning it as a considerable factor affecting the adoption/implementation of SOA in organisations. Almost all empirical studies stated that SOA governance was given top priority amongst all CSFs [11]. The researchers also claimed that to avoid failure in implementing SOA, governance mechanisms are needed. Otherwise, the architecture would be complicated, uncontrolled, fragile and would finally be discarded [11,31,51,52]. This result is in line with the outcome of Fig. 2, which shows that governance is one of the most frequently used terms in the primary papers analysed in this review study. Moreover, based on the findings presented in this section, Education and Training, Financial Issues/Costs, and SOA Strategy were the next important factors that were mentioned

12 times as significant elements in implementing SOA. Moreover, culture was considered as the most critical factor in different fields of study [53]. Additionally, most researchers and practitioners cited a lack of available skills as a concern in adopting SOA [54]. Similarly, the Organisational Culture factor, as well as Skill Availability, were identified as the next most influential factors in this systematic review. Also, the result shows that academic researchers determined the influential impact of SOA Team, Complexity, Communication, and Clear Goal Setting in implementing SOA in organisations, ranked as the ninth most important factor in the primary studies as whole. In addition, Roles and Responsibilities, SOA Policies, and Top Management Support were mentioned in 8 primary studies as significant enablers for adopting SOA. The rest of the factors that have been mentioned more than 2 times are Best Practices, Business And IT Alignment, Business Process, Clear SOA Project Roadmap, Compatibility, Enterprise Architecture, Experience, Expertise Availability, Immature Systems/Tool, IT and SOA Awareness, IT Infrastructure, Long-Term and Step-by-Step Transition Planning, Management Fashion, Organisational Alignment, Perceived Business Value, Project Identification, Resistance to Change, Resources, Reuse Capability, Risk, Security Concerns, Service Portfolio, SOA Business Capabilities, Standards, Strategic Alignment, Testing Challenges, and Vendor Support.

RQ (4): What are the gaps, limitations, and future work recommendations in the current researches of SOA?

In this systematic study, a large-scale literature review was conducted to determine the current state of SOA research, as SOA has proven to be a good comprehensive architecture for developing and implementing new technologies like the Internet of Things. However, the existing body of knowledge regarding the implementation of SOA is still limited.

Many researchers have expressed that organisations prevent the adoption of SOA due to difficulties in realising the full benefits of SOA, mainly because of SOA complexity relating to the nature of the services and the adoption levels [11,55]. The results of this study show that almost all previous studies focused on SOA at the organisational level without considering the individual perspective of adopting/implementing this technology. So, investigations into the adoption behaviours of individuals in implementing SOA are still inadequate, but many researchers stated that human factors have a significant influence on the adoption of SOA [56]. There are not enough empirical studies that have examined these factors from the perspective of top managers in organisations. Hence, this study recommends future empirical research to investigate the intention of top managers, decision-makers, and employees in implementing and adopting SOA in their organisations.

In a review paper, Pulparambil and Baghdadi [57] expressed that although many SOA maturity models have been proposed in recent years, there are still limited empirical investigations on the validation of these models. Since the selection of an appropriate SOA maturity model is a challenging issue for SOA experts and practitioners, more research is needed to assess existing SOA maturity models.

Moreover, a vast number of research studies demonstrated that applying IS models and frameworks indeed facilitates the process of adopting and implementing new technologies in enterprises [58,59]. Cho et al. [60] emphasised that it is crucial to ground the study design on pertinent models and theories to guarantee the effectiveness of the technology. Hence, it is recommended that the effectiveness of implementing SOA be investigated based on the relevant theories in an organisational setting.

In addition, user acceptance of new technologies and systems is considered essential for the successful implementation of any

Table B.1

Study ID	Full name of the journals, conferences, books source
S37	ACM Transactions on Management Information Systems (TMIS)
S13	AMCIS Proceedings
S19	Australasian Conference on Information Systems (ACIS)
S18, S15	Business & Information Systems Engineering
S28, S92	Business Process Management Journal
S8	Communications in Computer and Information Science
S63	Data & Knowledge Engineering
S95	Enterprise Information Systems
S32	European Conference on Information Management and Evaluation, ECIME
S2, S30, S76, S43	European Conference on Information Systems (ECIS)
S4, S11	European Conference on Software Maintenance and Reengineering
S77	Iberian Conference on Information Systems and Technologies (CISTI)
S80	IFIP International Conference on Advances in Production Management Systems
S21	Information and Software Technology
S24, S40	Information Systems and e-Business Management
S36	Information Systems Development
S57	Information Systems Frontiers
S9	Information Systems Management
S44	Information Technology for Development
S42	International Conference of Manufacturing Engineering Society, MESIC
S78	International Conference on Applied Science and Technology (ICAST)
S66	International Conference on Bioinformatics and Biomedicine (BIBM)
S56	International Conference On Computer and Information Sciences (ICCOINS)
S94	International Conference on Computer and Technology Applications
S25	International Conference on Contemporary Computing
S14	International Conference on Digital Content, Multimedia Technology and its Applications (IDCTA)
S12	International Conference on Digital Ecosystems and Technologies (DEST)
S70	International Conference on E-Health and Bioengineering Conference (EHB)
S82	International Conference on Electronics, Computers and Artificial Intelligence (ECAI)
S62	International Conference on Emerging Technologies and Innovative Business Practices for the Transformation of Societies (EmergiTech)
S79	International Conference on High Performance Computing and Communications, International Conference on Smart City and International Conference on Data Science and Systems
S46	International Conference on Industrial Applications of Holonic and Multi-Agent Systems
S16	International Conference on Information and Management Engineering
S23, S60	International Conference on Information Systems
S10	International Conference on Information Technology Interfaces (ITI)
S54	International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)
S53	International Conference on Information, Intelligence, Systems and Applications (IISA)
S74	International Conference on Mobile Web and Information Systems
S89	International Conference on Mobile Web and Intelligent Information Systems
S73	International Conference on Research and Innovation in Information Systems (ICRIIS)
S67	International Conference on Services Computing
S88	International Conference on Service-Oriented Computing
S87	International Conference on Technical Debt
S55	International Enterprise Distributed Object Computing Workshop
S26	International Joint Conference on Computer Science and Software Engineering (JCSSE)
S39, S59	International Journal of Business Information Systems
S101	International Journal of Computer Science and Network Security (IJCSNS)
S102	International Journal of Computing Sciences Research
S48	International Journal of Healthcare Information Systems and Informatics
S7	International Journal of Information Technology and Management
S84, S91	International Journal of Information Systems in the Service Sector
S31	International Journal of Intelligent Information Technologies
S69	International Journal of Production Research
S97	International Journal of Scientific and Technology Research
S98	International Journal of Web Information Systems
S27	International Symposium on Web Systems Evolution (WSE)
S17	International Systems Conference (SysCon)
S5	International Workshop on Enterprise Applications and Services in the Finance Industry
S85	Journal of Advanced Transportation
S90	Journal of Banking Technology (IDRBT)
S33, S34, S103	Journal of Enterprise Information Management
S93	Journal of Information and Communication Technology
S51	Journal of Information Technology Case and Application Research
S100	Journal of Intelligent Manufacturing
S64	Journal of King Saud University – Computer and Information Sciences
S49	Journal of Management Information Systems
S20	Journal of Software: Evolution and Process
S1	Journal of Theoretical and Applied Electronic Commerce Research (JTAER)
S83	Journal of Telecommunication, Electronic and Computer Engineering
S71	Kybernetes
S50	Managing in Recovering Markets
S68	Multi-Disciplinary Engineering for Cyber-Physical Production Systems

(continued on next page)

Table B.1 (continued).

Study ID	Full name of the journals, conferences, books source
S61	Pacific Asia Conference on Information Systems (PACIS)
S22	Pacific Asia Journal of the Association for Information Systems
S41	Procedia Technology
S29	Project Management Techniques and Innovations in Information Technology
S86	SAGE Open
S35, S99	Science and Information Conference (SAI)
S6, S58, S72	Service Oriented Computing and Applications
S65	Students' Conference on Electrical, Electronics and Computer Science (SCEECS)
S81	The Computer Journal
S47	The Handbook of Service Innovation
S38	The Journal of Strategic Information Systems
S45	Transforming Government: People, Process and Policy
S75	Twenty First Pacific Asia Conference on Information Systems
S52	Twenty-first Americas Conference on Information Systems
S3	Workshop on E-Business
S96	World Congress on Medical and Health Informatics, MEDINFO

IT project [61]. To achieve effective technological implementation and to reduce unexpected behaviour in adopting technologies, enablers, obstacles, and all issues related to the rejection of new technologies must be understood clearly [59,62]. However, the findings of this systematic review show that there are not enough empirical studies that have applied IS frameworks to examine the significant factors that affect the implementation of SOA in organisations. Some frameworks that could be applied are the Technology Acceptance Model (TAM), the Unified Theory of Acceptance and Use of Technology (UTAUT), the Theory of Planned Behaviour (TPB), the Theory of Reasoned Action (TRA), and the Technology-Organisation-Environment (TOE) Framework. Hence, it is recommended that future works empirically examine SOA-related factors based on the IS frameworks mentioned above. The researcher has proposed an integrated framework in another study to cover this research gap [49]. This proposed model could serve as a starting point for future works.

However, from the methodological perspective, only a few of the primary studies used design science and mixed-method methodologies. The researchers also found that many researchers believed that the mixed method (both quantitative and qualitative) could explain the research problem more comprehensively rather than only using the quantitative or qualitative method separately [63]. Hence, further studies should apply design science and mixed method techniques to offer more value to the SOA research domain. Similarly, most of the previous studies applied case studies, but other research strategies could offer more valuable insight to guide the development of SOA in organisations. Hence, it is suggested that future works conduct large-scale surveys, as well as other research strategies such as experiments, action research, and archival research.

Academics have claimed that integrating SOA and other technologies could assist organisations to remain relevant in today's competitive scene, besides granting many other organisational benefits [5,64]. For example, Fischer et al. [65] explored the combination of business process management, business rules management, and SOA to facilitate business responsiveness, operational agility, and quality. Their results showed that actual practice and investigation lacked concepts and approaches that enabled organisations to understand the benefits of integrated solutions. Concerning this issue, the findings of the current SLR confirm that not much research has been done on the integration of SOA and other technologies to provide a unified solution for organisations. Despite the trend towards SOA adoption principles in the Internet of Things field is reported by many academics [4,66], the results of this review show that the existing literature is still limited in this domain.

Furthermore, academics emphasised that the cost estimation of SOA is one of the critical challenges for organisations. For

example, Alanazi et al. [67] indicated that estimating the cost and effectiveness of implementing SOA is a highly complex issue since this estimation needs the enterprise service bus (ESB) evaluation within the systems. The existing cost estimation approaches of SOA-based systems are insufficient [67]. Consequently, more research is needed to evaluate the cost estimation and effectiveness of SOA-based systems.

6. Conclusion

A brief overview of previous SOA-related papers highlighted the absence of a detailed systematic literature review (SLR) that has extracted information about SOA research strategies, methodologies, theories, models, and/or influential factors. Therefore, this study conducted an SLR to extract all potential factors affecting SOA implementation from 2009 to November 2019. To the best of the authors' knowledge, this is the first SLR study that has conducted an in-depth analysis from an IS standpoint. This review included the IS domain in the search because SOA is predominantly adopted in IS. It should be noted that since this study was recently carried out in 2019, the authors could not guarantee the inclusion of all publications in 2019 in the SLR. Through extensive search, 103 papers were selected as the primary studies for the SLR. The primary papers were then classified into 4 themes, namely SOA Adoption, SOA Impact, SOA Concept, and SOA Practice. Due to a lack of space to accommodate the total number of factors (more than 200), this study only investigated 41 factors as the most important ones, as these were repeatedly perused at least 3 times in previous studies. Of these factors, based on the analysis, SOA Governance appeared to be the most important factor because it was cited 30 times as a significant factor for adopting/implementing SOA.

Based on the results, the majority of the primary studies applied the qualitative research method (40%) followed by the quantitative method (28%). Finally, the existing research gaps and limitations of SOA research were discussed, and some future directions and recommendations for academic work provided. In conclusion, since SOA is the main foundation of the world's leading technologies, this systematic review could prove valuable for both academic researchers and SOA experts. Furthermore, this study established a mind map of SOA themes, theories, and methodologies for academic researchers to quickly hone in on a research area that requires more investigation. Also, the results of this study would help top managers and decision-makers to understand the most important factors affecting the implementation of SOA in their organisations. Furthermore, the findings of this review would also assist academics, especially novice researchers, to realise the existing limitations and gaps, as well as recommended future works, in this field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Primary studies

See Table A.1.

Appendix B. Name of journals, conferences, and books sources

See Table B.1.

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